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Tech Pulse FY– 26

In-Vehicle Networking: Assessment Plan (University Implementation Guide)

Tech Pulse: In-Vehicle Networking Assessment Plan

I. Overall Assessment Strategy

1. TTL Role

- Provide standardized IVN Capstone project Rubrics
- Perform validation of evaluated submissions using the same rubric
- Moderate scores if deviation exceeds $\pm 10\%$

2. University Role

- Conduct evaluation of IVN Capstone Projects using standard rubrics
- Maintain academic integrity and verify original work

3. Student Role

- Submit IVN project artifacts: DBC File, Panel Design, CAPL Scripts, Simulation Setup, Documentation, and Video.
- Ensure proper folder structure and naming conventions.

II. Student Submission Requirements

1. Technical Artifacts

Each student must submit:

- DBC File(s) – signals, messages, nodes, attributes.
- Panel Design file (with relevant panel controls and panel elements)
- CAPL Scripts – message handling, automation, test logic.
- CANoe configuration file

- Simulation Logs – BLF/ASC format
- Optional: Additional automation scripts, Python helpers, custom functions.

2. Technical Report (PDF)

Must include

- Problem statement
- DBC file description
- Automotive Communication Network architecture
- Panel layout description
- CAPL logic explanation & flow-diagrams
- Simulation steps & results
- Innovation highlights (custom CAPL, dynamic panels, etc.)
- Screenshots, logs, and evidence
- Reflection & conclusions

3. Video Demonstration (5–10 minutes)

- Walkthrough of DBC file, panel dashboard, CAPL, integration, simulation, and results.
- Real-time demonstration of system behavior.

III. Evaluation Rubrics (100 Marks)

Rubrics for evaluating Capstone project for In-Vehicle Networking course

Criteria	Excellent	Good	Fair	Poor
1. DBC File Creation (20 Marks)	Accurately creates a complete DBC file with message IDs, signal definitions, bit positions, data lengths, scaling, offsets, units, node assignments, and attributes. DBC validates successfully and supports intended CAN communication. (17-20) marks	Creates a mostly correct DBC file with minor errors in signal configuration or metadata. Communication functionality remains largely unaffected. (13-16) marks	DBC file contains multiple configuration errors affecting message interpretation or communication. Partial implementation achieved. (8-12) marks	DBC file is incomplete, invalid, or contains major errors preventing proper CAN communication. (0-7) marks
2. Panel Design (20 marks)	Professional, intuitive, and functional panel layout; all required panel controls and panel elements	Functional panel with minor design flaws;	Basic panel with limited functionality;	Panel missing or non-functional. (0-7) marks

	implemented; responsive design. (17-20) marks	most elements implemented. (13-16) marks	lacks clarity or usability. (8-12) marks	
3. CAPL Scripting (20 marks)	CAPL scripts are optimized, well-commented, and achieve all required automation and logic; error-free execution. (17-20) marks	CAPL scripts functional with minor inefficiencies or limited comments. (13-16 marks)	CAPL scripts partially functional; logic incomplete or contains errors. (8-12) marks	CAPL scripts missing or non-functional. (0-7) marks
4. Integration & Simulation (20 marks)	Seamless integration of DBC, panel, and CAPL; simulation runs without errors; results validated. (17-20) marks	Integration is mostly successful, minor simulation issues. (13-16) marks	Integration is incomplete; simulation runs with significant errors. (8-12) marks	No integration or simulation fails completely. (0-7) marks
5. Documentation & Explanation (10 marks)	Clear, detailed documentation of steps, rationale, and results; includes screenshots and flow diagrams. (9-10) marks	Documentation present but lacks depth or clarity in some areas. (7-8) marks	Minimal documentation; missing key details. (4-6) marks	No documentation or extremely poor explanation. (0-3) marks
6. Innovation & Problem-Solving (10 marks)	Demonstrates creative solutions, advanced features (e.g., custom CAPL functions, dynamic panels). (9-10) marks	Some creative elements; mostly standard implementation. (7-8) marks	Limited creativity; basic implementation only. (4-6) marks	No creativity; incomplete or copied work. (0-3) marks

IV. Overall Weightage Structure

Component	Weightage
MCQ	60%
Capstone Project (As per the Rubric)	40%

V. Faculty Process: Synopsis, Score Consolidation & Validation

1. Synopsis Stage

Students submit and Faculty validates technical relevance.

- Project Title/ Topic
- Subtopic
- Scope
- Expected outcomes

2. Score Consolidation

- Evaluate using the 6-criterion, 100-mark rubric.
- Fill master score sheet → UniversityName_IVN_FinalScores.xlsx.
- Attach evidence (logs, DBC, panel, CAPL, video).

3. TTL Validation & Moderation

- TTL evaluates submissions.
- If deviation $> \pm 10\%$ → reevaluation/moderation triggered.
- Final score approved by TTL.

VI. Folder Structure for Final Submission

UniversityName_StudentName_RegisterNo_IVN/

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├── DBC/
├── Panel/
├── CAPL/
├── Integration/
├── Logs/
├── Documentation/
├── Video/
└── ZIP of entire folder
  
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VII. Assessment Timeline

- Week 1 – Guidelines + Synopsis
- Week 2 – Faculty Review
- Week 3 – IVN MCQ Assessment

- Weeks 3–5 – IVN Project Development
 - Week 6 – Capstone Project Evaluation
 - Week 7 – TTL Validation & Moderation
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